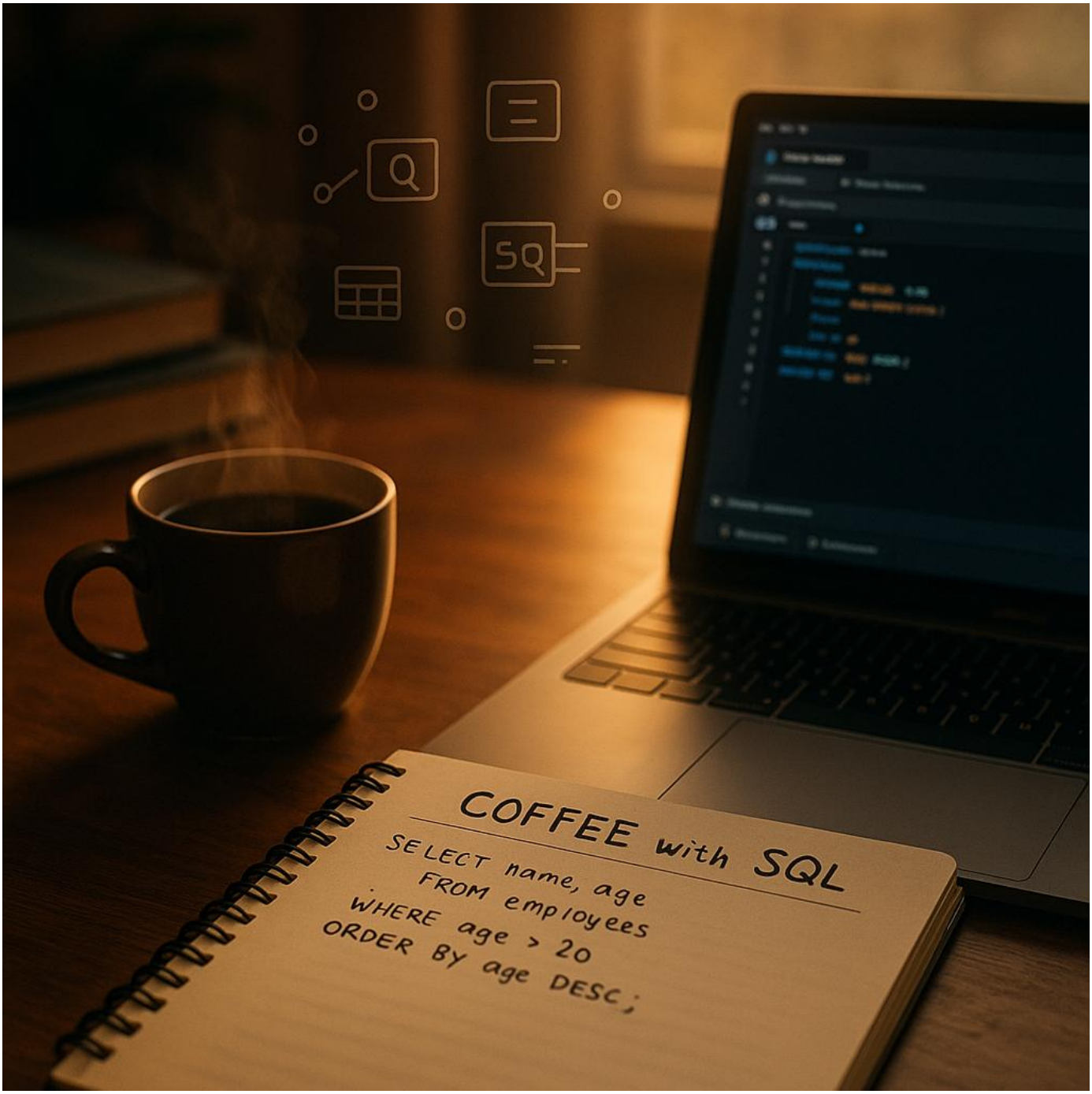
PROBLEM 7

Technical Interview Question

SQL & PySPARK - Solution



i/p:

	team_1 character varying (20)	team_2 character varying (20)	winner character varying (20)
1	India	SL	India
2	SL	Aus	Aus
3	SA	Eng	Eng
4	Eng	NZ	NZ
5	Aus	India	India

Desired o/p

	team character varying (20)	match_played bigint	matches_won bigint	matches_loss bigint
1	India	2	2	0
2	Eng	2	1	1
3	NZ	1	1	0
4	Aus	2	1	1
5	SA	1	0	1
6	SL	2	0	2

For Data Analyst: How would you solve this in **SQL**?

For Data Engineer: How would you solve this in **PySpark**?

Think a bit before scrolling…

SQL Soln.

Step-1: create a winner_flag

For Team_1

```
select
    team_1 as Team,
    case when team_1 = winner then 1 else 0 end as winner_flag
from icc_world_cup
```

For Team_2

```
select
    team_2 as Team,
    case when team_2 = winner then 1 else 0 end as winner_flag
from icc_world_cup
```

Step 2: Union the Team_1 & team_2

```
select
    team_1 as Team,
    case when team_1 = winner then 1 else 0 end as winner_flag
from icc_world_cup

union all

select
    team_2 as Team,
    case when team_2 = winner then 1 else 0 end as winner_flag
from icc_world_cup
```

Step 3 : Check-Mate

```sql
with flat_data as (
select
    team_1 as Team,
    case when team_1 = winner then 1 else 0 end as winner_flag
from icc_world_cup

union all

select
    team_2 as Team,
    case when team_2 = winner then 1 else 0 end as winner_flag
from icc_world_cup
)
select
    team,
    count(1) as match_played,
    sum(winner_flag) as matches_won,
    (count(1) - sum(winner_flag)) as matches_loss
from flat_data
group by team
order by 3 desc
```

PySpark Soln.

Prerequisite: Prepare the data

```
data = [
    ('IN', 'SL', 'IN'),
    ('SL', 'Aus', 'Aus'),
    ('SA', 'Eng', 'Eng'),
    ('Eng', 'NZ', 'NZ'),
    ('Aus', 'IN', 'IN')
]
colname = ["team_1","team_2","winner"]

df = spark.createDataFrame(data, schema=colname).repartition(8)
```

#repartition is optional

Step 1: Prepare team1_data

```
team1_data = (
    df.withColumn( colName: "Team", col("team_1"))
    .withColumn( colName: "winner_flag", expr('''case
                                                    when
                                                        team_1 = winner
                                                    then 1 else 0
                                                    end'''))
    .select("Team", "winner_flag")
)
```

Step 1: Prepare team2_data

```
team2_data = (
    df.withColumn( colName: "Team", col("team_2"))
    .withColumn( colName: "winner_flag", expr('''case
                                                    when
                                                        team_2 = winner
                                                    then 1 else 0
                                                    end'''))
    .select("Team", "winner_flag")
)
```

Step 3: Union all the data

```
union_data = team1_data.union(team2_data)
```

Step 4: Check-Mate

```
finaldata = (
    union_data.groupby("Team")
    .agg(
        count("Team").alias("matches_played"),
        sum("winner_flag").alias("matches_won")
    )
    .withColumn( colName: "matches_lost", col("matches_played") - col("matches_won"))
    .orderBy(col("matches_won").desc())
)
finaldata.show()
```